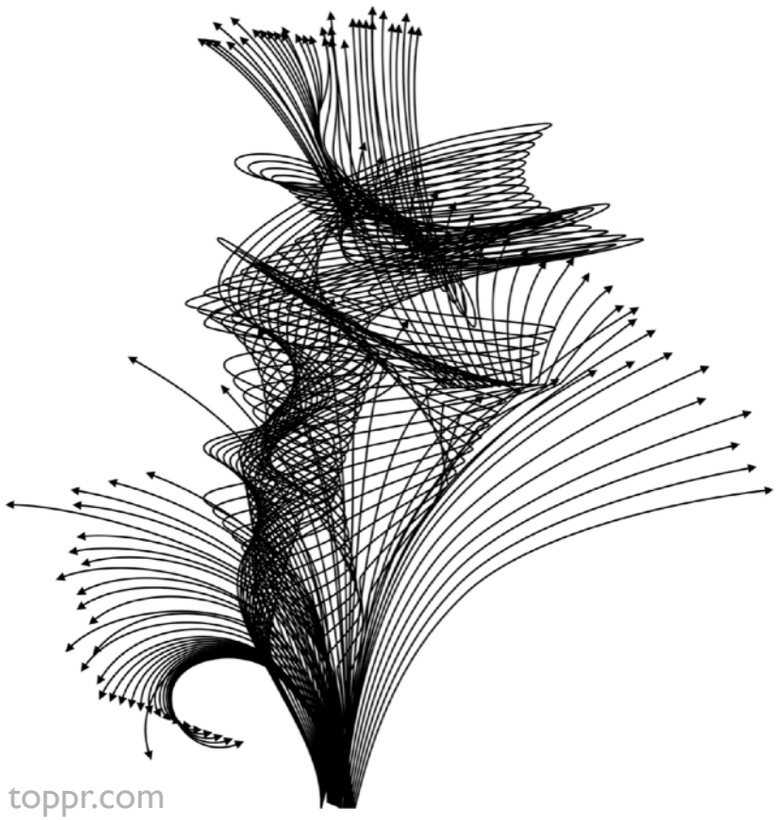
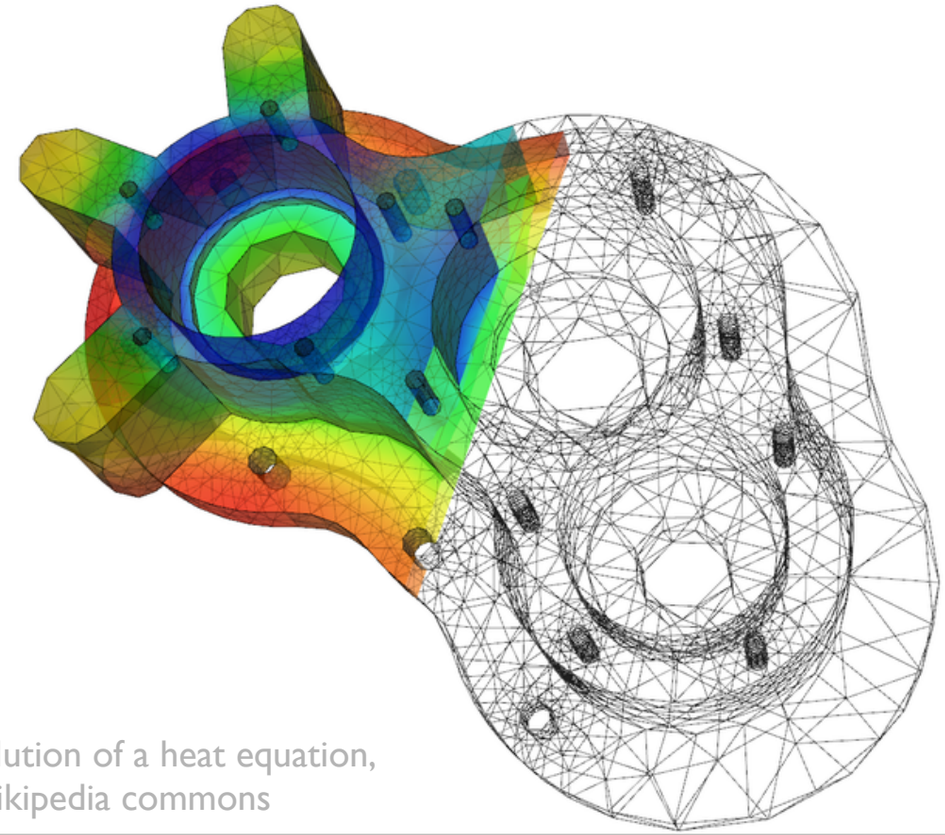
Week 11 - #1

Ordinary Differential Equations (III)



toppr.com



Solution of a heat equation,
Wikipedia commons

Today: Ch 8

Next Class: Ch 8

Ji-hoon Kim (Seoul National University)

This material is for the educational purpose only, and is not for public consumption or mass distribution without the author's explicit consent.
Despite his best efforts, the author apologizes for any inaccurate or outdated information and references.

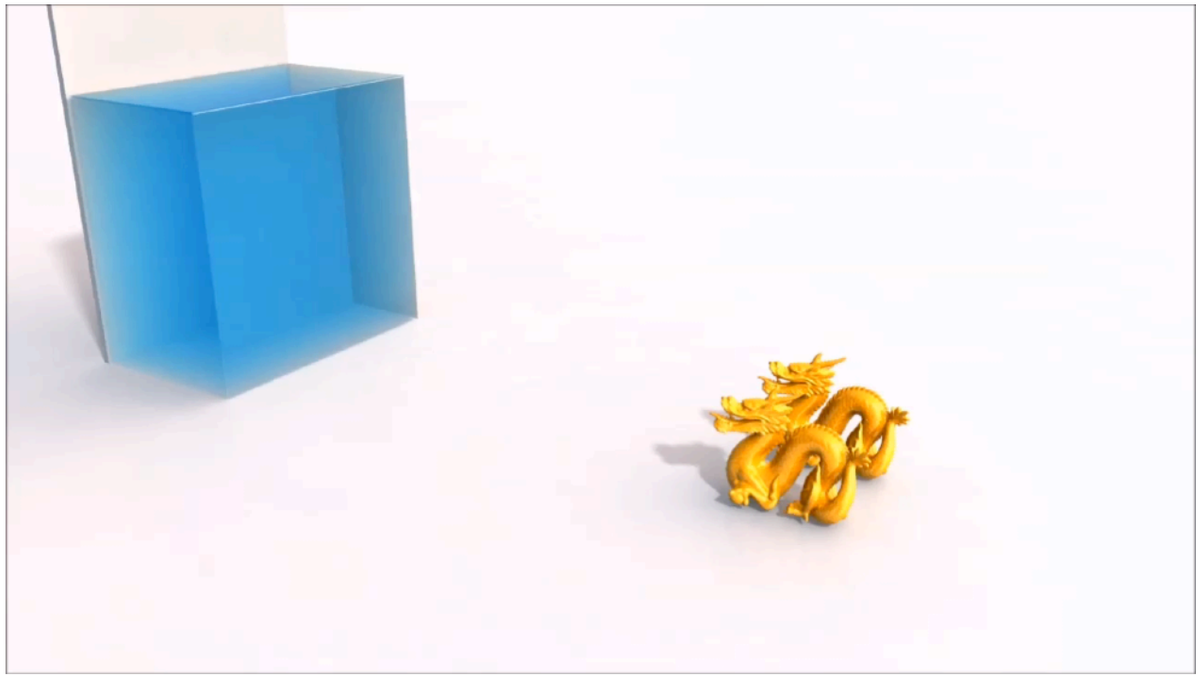
Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #16

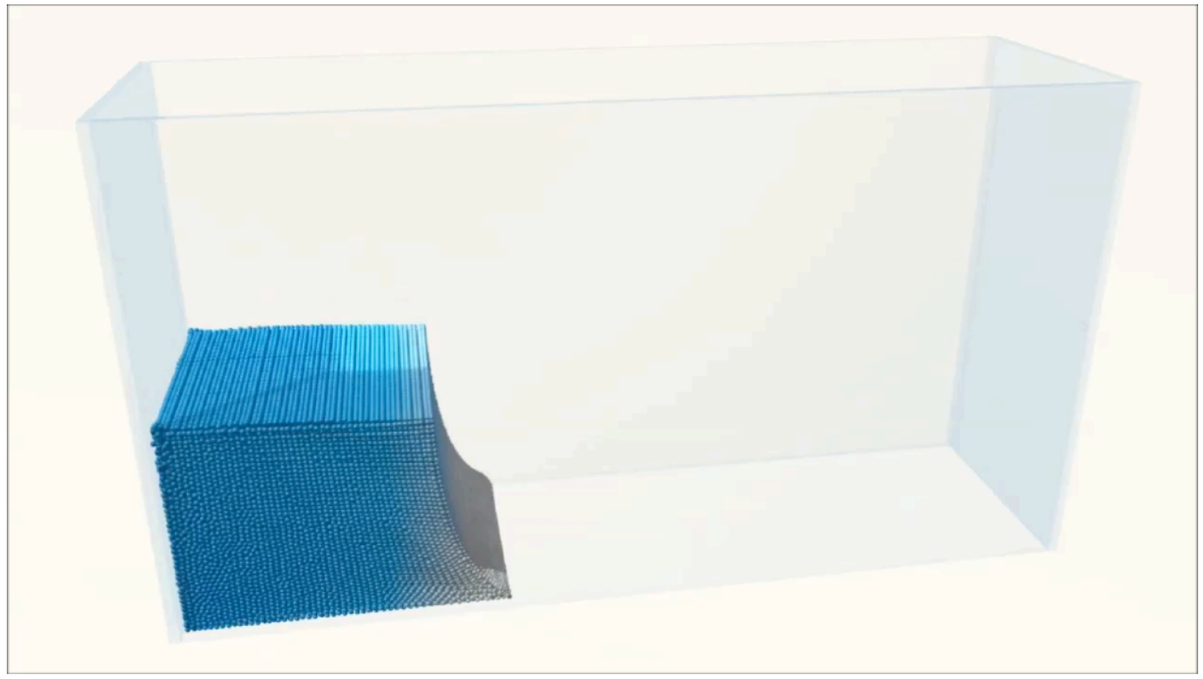
— [open book and open note, but no cellphone or laptop, drop it off as you leave the class] —

Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Boas, Chapter 8, Section 5, Problem 5

2. Briefly explain, without using detailed math, what the movies below try to visualize. How do you think these movies were made?





Computational fluid dynamics, Bender & Koschier, www.youtube.com/watch?v=POnmzzhc5E0

Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 8, Boas, 3rd ed.

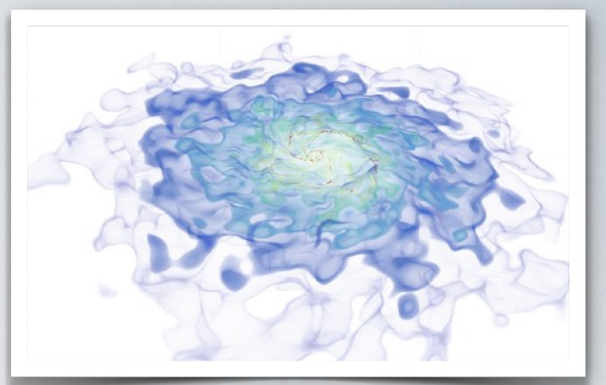
The problems I suggest you to take a deeper look into include, but are not limited to, the following. The class homework assignments will mainly be from this list.

- Section 01: Problems 4
- Section 02: Problems 6, 8, 11, 12, 20(b), 34
- Section 03: Problems 4, 11, 14, 16, 20
- Section 04: Problems 2, 6, 8, 10, 15, 20
- Section 05: Problems 6, 9, 11, 13, 15, 24, 31(c)(d)
- Section 06: Problems 5, 8, 10, 14, 21, 23, 28, 34, 35, 42
- Section 07: Problems 2, 3, 8, 17, 22, 30
- Section 08: Problems 1, 2, 4, 7, 10, 11, 12, 19, 22, 26
- Section 09: Problems 8, 9, 14, 16, 28, 30, 34, 38
- Section 10: Problems 3, 8, 11, 14, 18
- Section 11: Problems 6(b)(c), 7, 9, 15, 21, 23, 25
- Section 12: Problems 2, 4, 6, 11, 16

<u>Chapter 13, Sections 1-4</u>

- Section 01: Problems 1, 4
- Section 02: Problems 1, 2, 3, 7
- Section 03: Problems 2, 3, 7
- Section 04: Problems 2, 5

76 problems / 383 total



Check your midterm score by Nov. 10

(Grader TAs: p1, 2, 6 - 서현우, hyeonu@snu.ac.kr
p3, 4, 5 - 신찬미, chanmi0412@snu.ac.kr)

(Mean: 13.67 / Standard Deviation: 4.49)

(Claim hour: Nov. 9, 17:30-19:00, 56-322)